

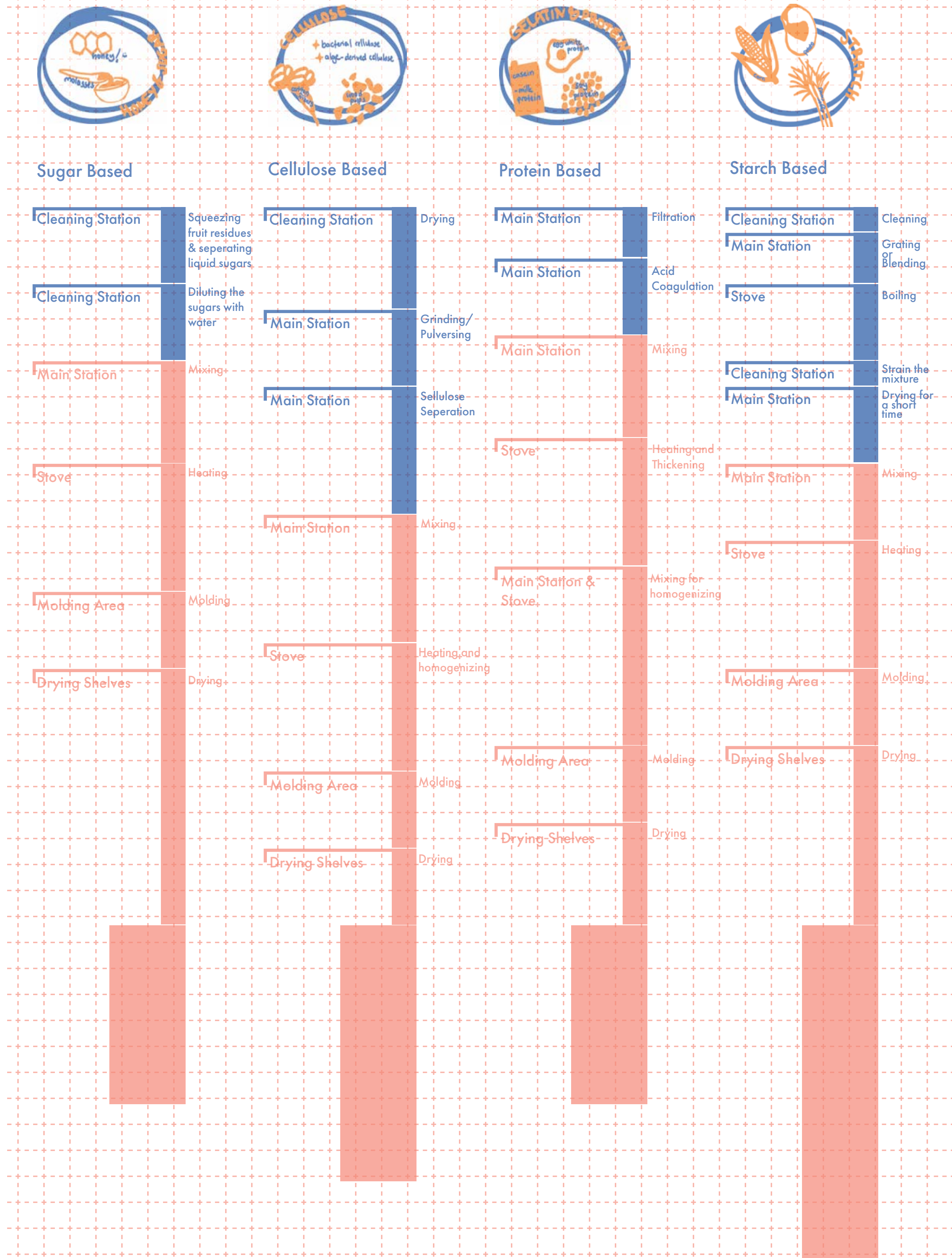
BIO DECAY

There are microplastics in kind of everything that we eat. This information theoretically means that we can make bioplastics out of everything that has microplastics in it.

In the Milli Reasürans building, there are many places for eating and drinking. These places generate a significant amount of food waste every day, and some of the nutrients derived from this waste have the potential to serve as materials for bioplastic production.

External factors, including humans, leave traces on the environment, whether intentionally or unintentionally. I archive these traces through bioplastics. In doing so, I both promote the use of bioplastics over petroleum-based plastics and document the micro-level effects these influences have on bioplastics by experimenting with their decay and wear patterns, thus, creating material for the archive.

How to make bioplastics?

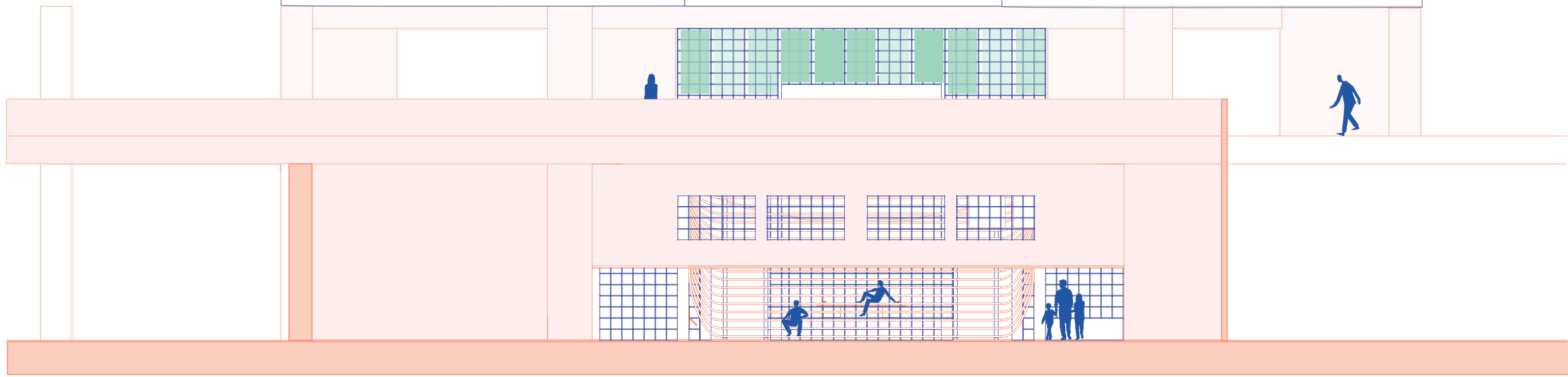
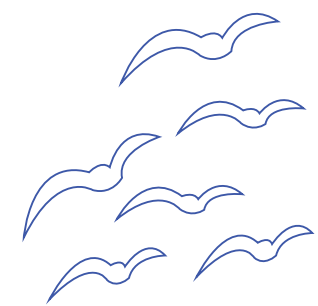
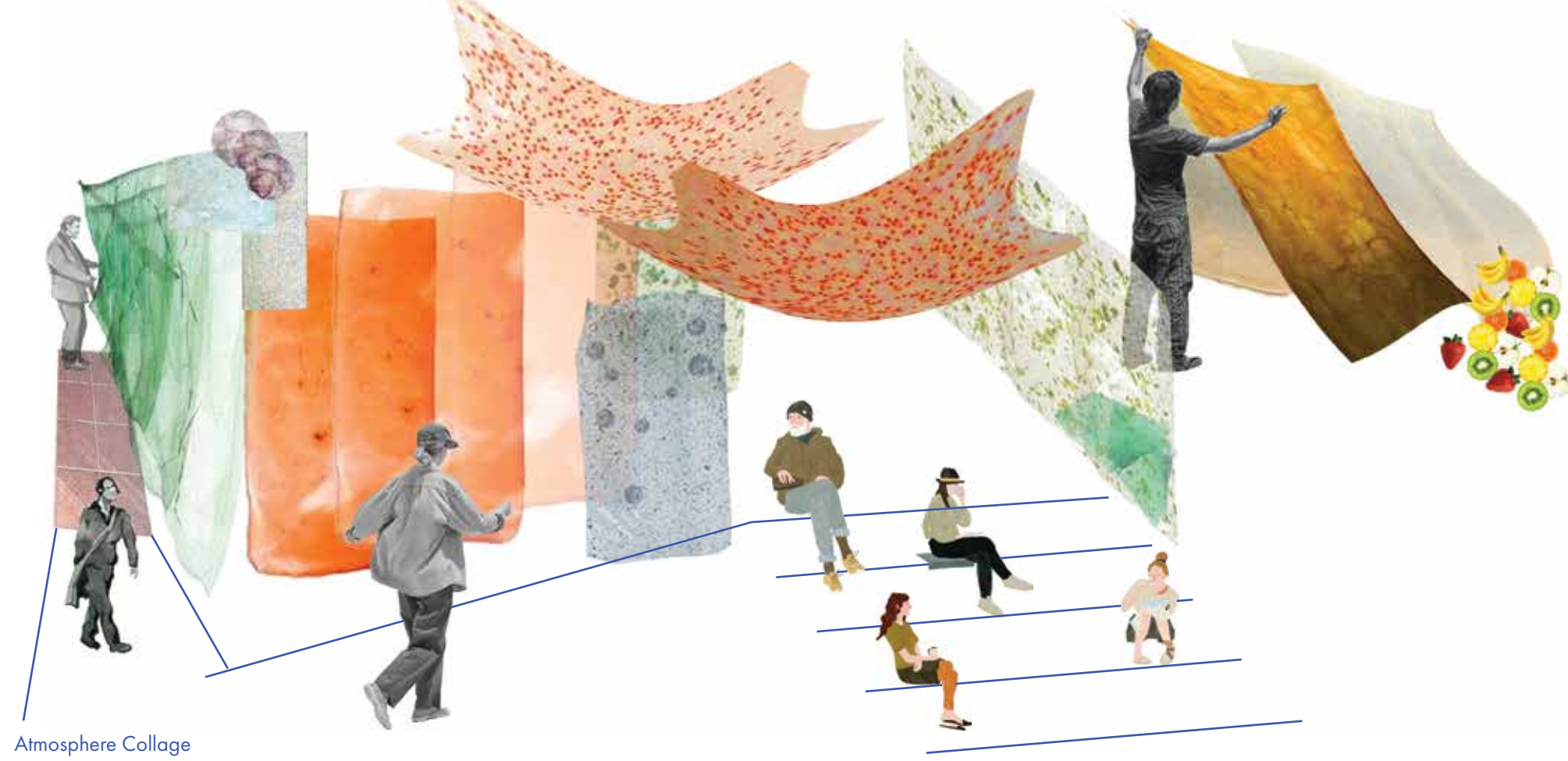


This project explores the relationship between **material decay, waste cycles, and spatial transformation** through the production and archiving of bioplastics. Located in the passage of the Milli Reasürans building, the project collects organic waste from nearby restaurants and from pedestrians consuming food in the area. These collected food scraps—such as fruit peels or vegetable residues—are categorized and processed on site to produce different types of bioplastics (starch-based, sugar-based, protein-based, cellulose-based).

Once produced, the bioplastics are shaped and hung on to the grid walls and custom-built wire structures by researchers working in the space. These bioplastics are not permanent; instead, they are designed to rot, crack, and change form over time. **The space becomes an archive—not to preserve but to document disappearance.** Each bioplastic piece is tagged and recorded, creating a timeline of material transformation.

As the bioplastics decay, the opacity levels of the surrounding architectural elements change as well. These shifts in transparency create dynamic light and shadow conditions in the space, subtly reflecting the degradation process. Thus, both the archive and the spatial atmosphere evolve together, emphasizing how human presence and environmental triggers like humidity and pollution interact with organic materials.

A living archive: fragile, temporary, and always in transformation.



1/100 Facade

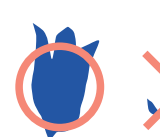
Common bioplastic decay process



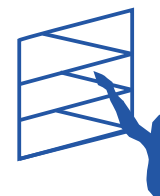
Regular Program



Collect food waste from restaurants and cafes nearby



Separating food wastes that can be transformed into bioplastics from those that cannot



Separating food wastes that can be transformed into bioplastics from those that cannot



Separating food wastes that can be transformed into bioplastics from those that cannot



As the bioplastic decays, its permeability first decreases, then increases. Its size gradually shrinks. The texture becomes harder and starts to lose flexibility. Its color also darkens over time.

As time passes, the opacity of the bioplastics changes. This also affects how the spaces are used. The function of the space transforms with the material. The change in opacity creates a natural timeline for how each zone is used.

In this project, the structure of the space is based on the grid structure. This grid works like a skeleton. On top of it, bioplastic materials are added like a second layer, like skin. These bioplastics are made from food waste collected from nearby restaurants and people using the area. The bioplastic layer is not permanent. Over time, it starts to decay, break, or change shape. When it changes, the skeleton underneath becomes visible again. This creates a space that is always transforming. It is never finished — it is always in progress. The changing material shows how time and human actions shape the space. The space becomes something alive, always growing, aging, and renewing itself.